



slides [tinyurl.com/icap2026-commitment](https://tinyurl.com/icap2026-commitment)

# *Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature*

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# Coauthors

The team behind this AI-assisted meta-analysis



**David J. Stanley**

University of Guelph



**Jose A. Espinoza**

Western University



**Amir Mehr**

The Synthesis Company

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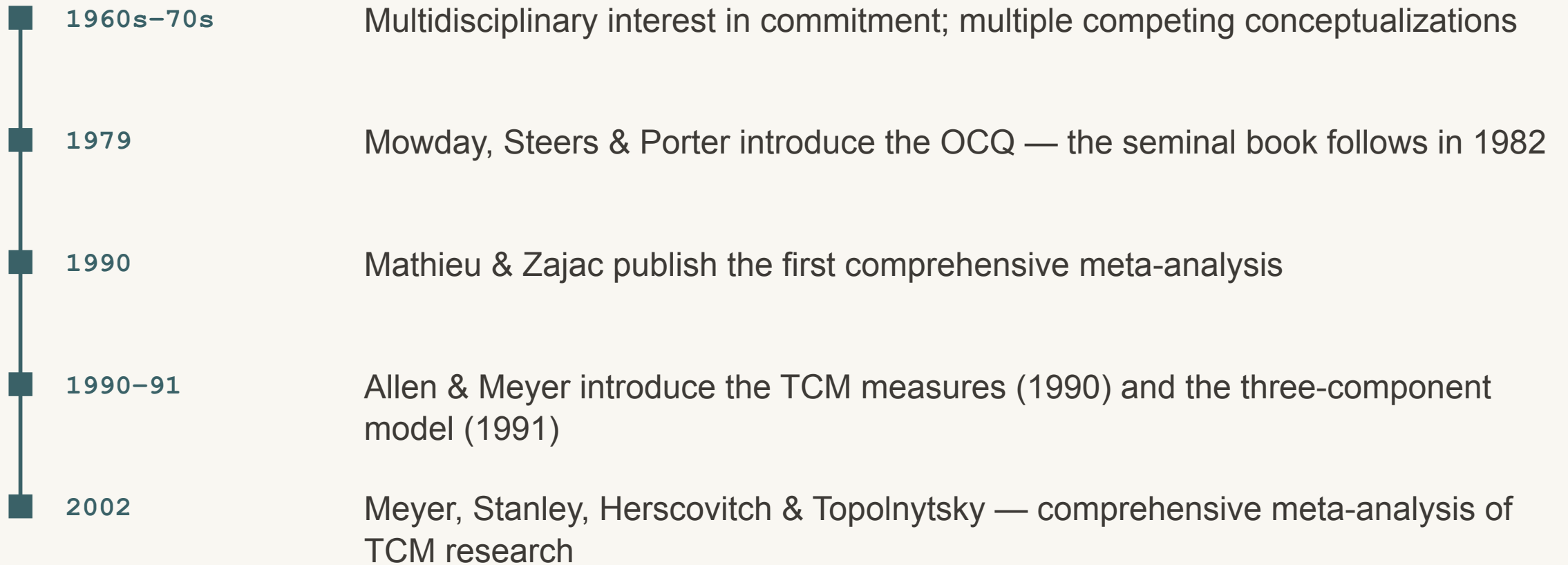
# Agenda

Where we've been, where we are, and where this is going

- Brief History of Commitment Theory & Research
- Challenges to conducting comprehensive reviews
- Toward an AI-Assisted Meta-Analysis
  - Where we are
  - Where we are going
  - What is possible

# Organizational commitment: the early years

A brief history — some milestones



# Two comprehensive meta-analyses

Studies (**k**) and participants (**N**) per analysis — Mathieu & Zajac (1990) and Meyer et al. (2002)

## 1990

Mathieu & Zajac (1990)

STUDIES PER ANALYSIS (**k**)

**3–43**

PARTICIPANTS PER ANALYSIS (**N**)

**302–15,531**

VARIABLES META-ANALYZED **48**



## 2002

Meyer et al. (2002)

STUDIES PER ANALYSIS (**k**)

**3–69**

PARTICIPANTS PER ANALYSIS (**N**)

**441–23,656**

VARIABLES META-ANALYZED **38**



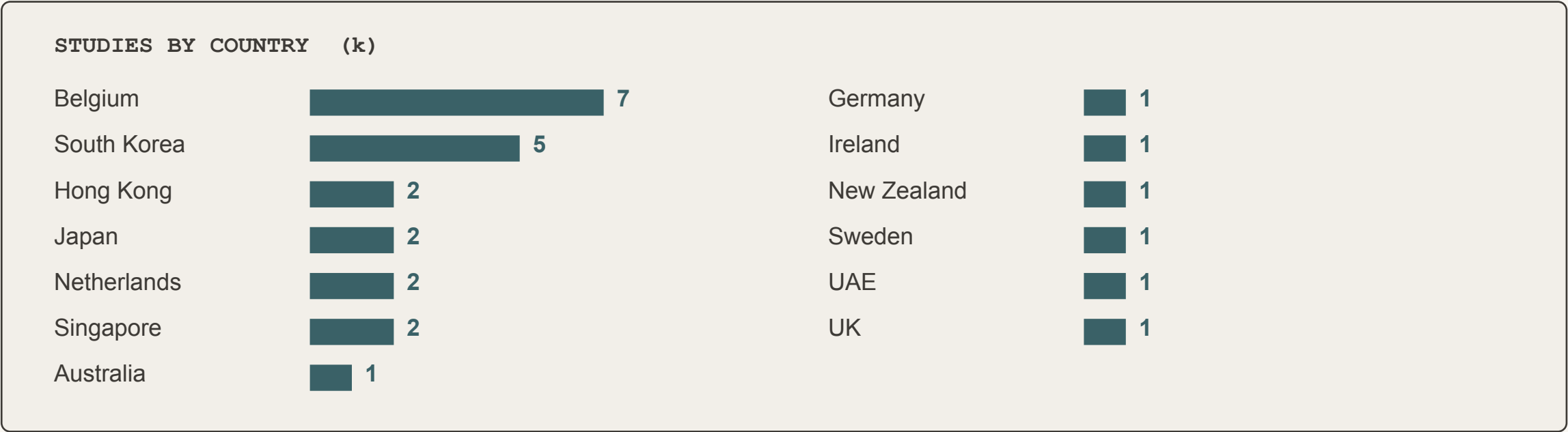
■ antecedents   ■ correlates   ■ consequences

# A modest cross-national base

Meyer et al. (2002) — studies reporting country of origin

■ GEOGRAPHIC REACH OUTSIDE NORTH AMERICA

13 countries · 27 studies



Studies reporting country of origin (n = 27); tabulation from the ICAP 2014 presentation.

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# Challenges to conducting comprehensive reviews

Why a truly comprehensive commitment review has become impractical by hand.

*1*

## **Studies keep growing**

Commitment research piles up faster every decade.

*2*

## **Reviews turn narrow**

Meta-analyses cover fewer variables to stay feasible.

*3*

## **Even narrow is too much**

Now even targeted reviews outstrip human capacity.

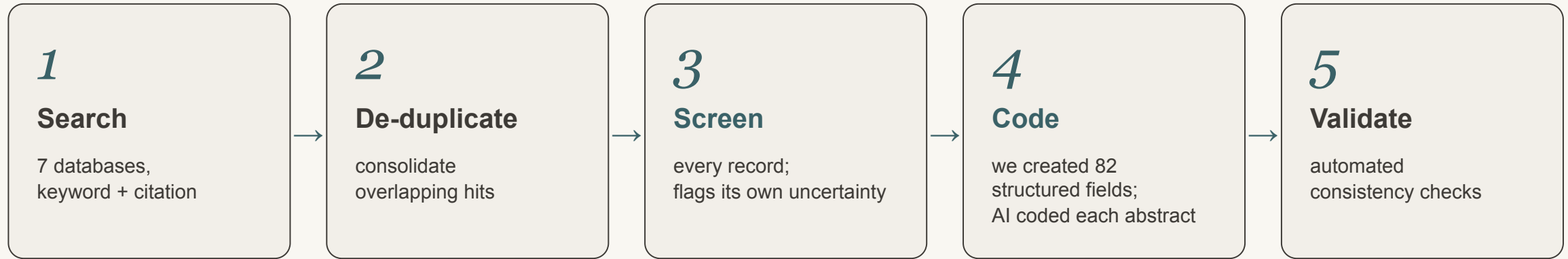
*4*

## **AI shares the labour**

AI screens and codes at scale, clearing the bottleneck.

# An AI-assisted pipeline — with humans in the loop

Screening and coding were performed by Eva, a family of AI agents; the team set the rules and decided every uncertain case.



## HUMANS STAY IN THE LOOP

- Set & refine the inclusion rules
- Decide every case the AI flags as uncertain
- Pilot-review the coding before each full run

The AI does the volume; the team sets the rules and owns the judgment calls.

FINDING THE STUDIES

# *Literature Search*

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A Global Search

# Literature search method

Two complementary search strategies in every database (except ERIC)

- **Keyword & subject-term search** — truncated organizational terms (org\*, work\*, employ\*, job\*, firm\*, institution\*, affect\*, norm\*, continu\*) within three words of commit\*, plus each database's subject headings; syntax adapted to each platform.
- **Cited-reference search** — every record citing any of 14 foundational commitment works (e.g., Porter et al., 1974; Mowday et al., 1979; Allen & Meyer, 1990; Meyer & Allen, 1991; Meyer et al., 2002; Klein et al., 2014).
- **Scholarly sources only** — academic journals, books, dissertations & theses, conference papers, reports, and working papers.
- **All languages** — the search applied no language restriction, so abstracts in any language were captured, not only English.

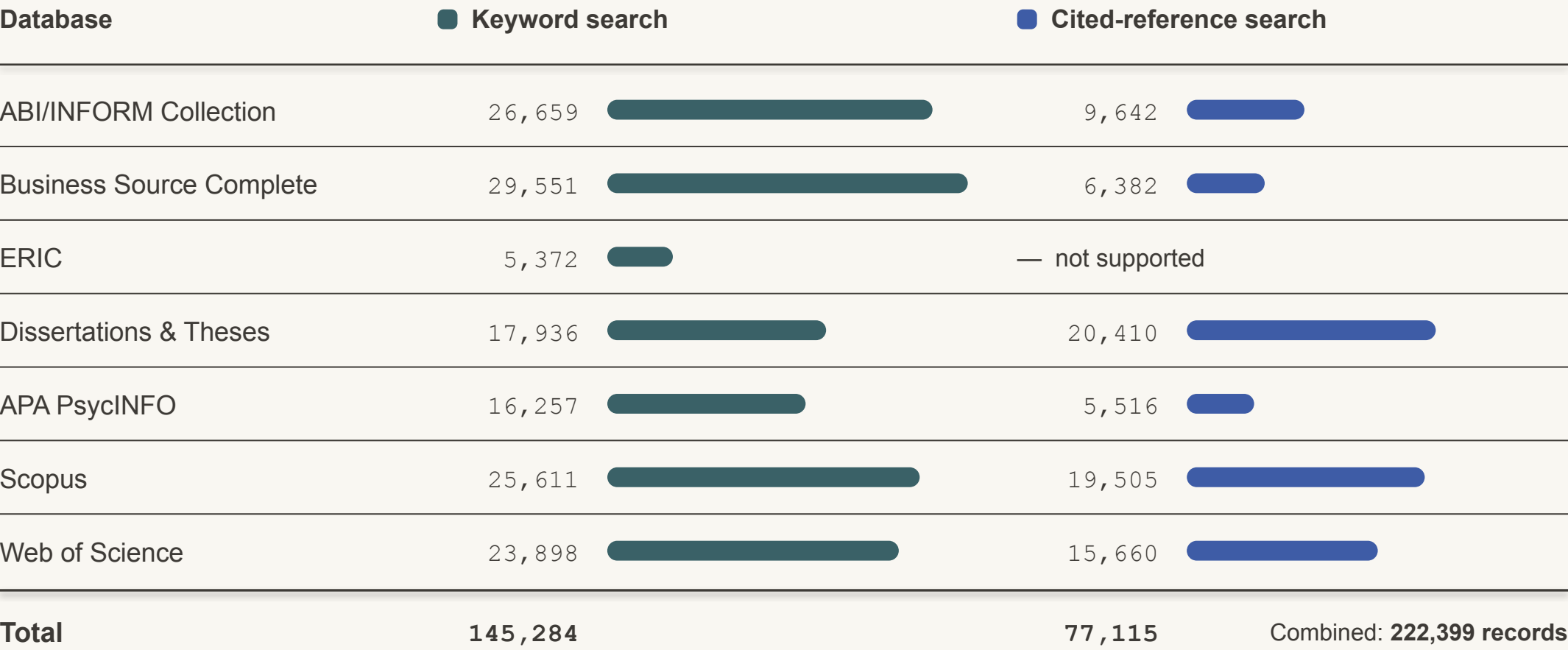
## Seven databases

Searched December 20, 2025 – January 7, 2026

|                          |                     |
|--------------------------|---------------------|
| ABI/INFORM Collection    | ProQuest            |
| Business Source Complete | EBSCO               |
| ERIC                     | Ovid · keyword only |
| Dissertations & Theses   | ProQuest            |
| APA PsycINFO             | Ovid                |
| Scopus                   | Elsevier            |
| Web of Science           | Clarivate           |

# Records identified, by database and search strategy

Records retrieved per database, before consolidation and de-duplication · bars share one scale



ERIC (Ovid) does not support cited-reference searching.



# Retrieved to Coded

De-duplication only removes overlap — AI relevance screening did the real narrowing

**222,399** records retrieved  
7 databases · keyword + citation



▼ de-duplication — *removes duplicate hits only*

**113,643** unique records  
after de-duplication



▼ AI relevance screening by the Eva family of agents — on each record's title & abstract

**22,484** coded abstracts  
22,605 relevant;  
121 had no abstract to code



Only ~20% of unique records passed AI relevance screening (22,605 of 113,643) — the rest were off-topic, not duplicates.

# How accurate is AI screening? — from prior research

## ■ Relative Screening Accuracy

**97.2%** **91.6%**  
AI expert humans

of relevant studies kept in rather than missed  
5 meta-analyses · 30,150 abstracts screened

## ■ Time saving with AI

**≈ 72×**

faster than screening by hand

36 hours of hand screening, done by the AI in under 30 minutes.  
One of the five · 2 reviewers + 1 adjudicator · AI: a single pass

## ■ How the accuracy was measured

The AI and the human team each screened every title and abstract, deciding relevant or not. Where the two disagreed, the record was re-checked to establish which studies genuinely belonged. The figure shown is the percentage of those relevant studies a method correctly kept in rather than missed, averaged across the five datasets.

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[Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. \(2026\). \*Improving Evidence Synthesis with Artificial Intelligence\* \[Preprint\].](#)

WHAT THE CODED ABSTRACTS REVEAL

# *Abstract Analysis*

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How the literature has grown — and what it studies.

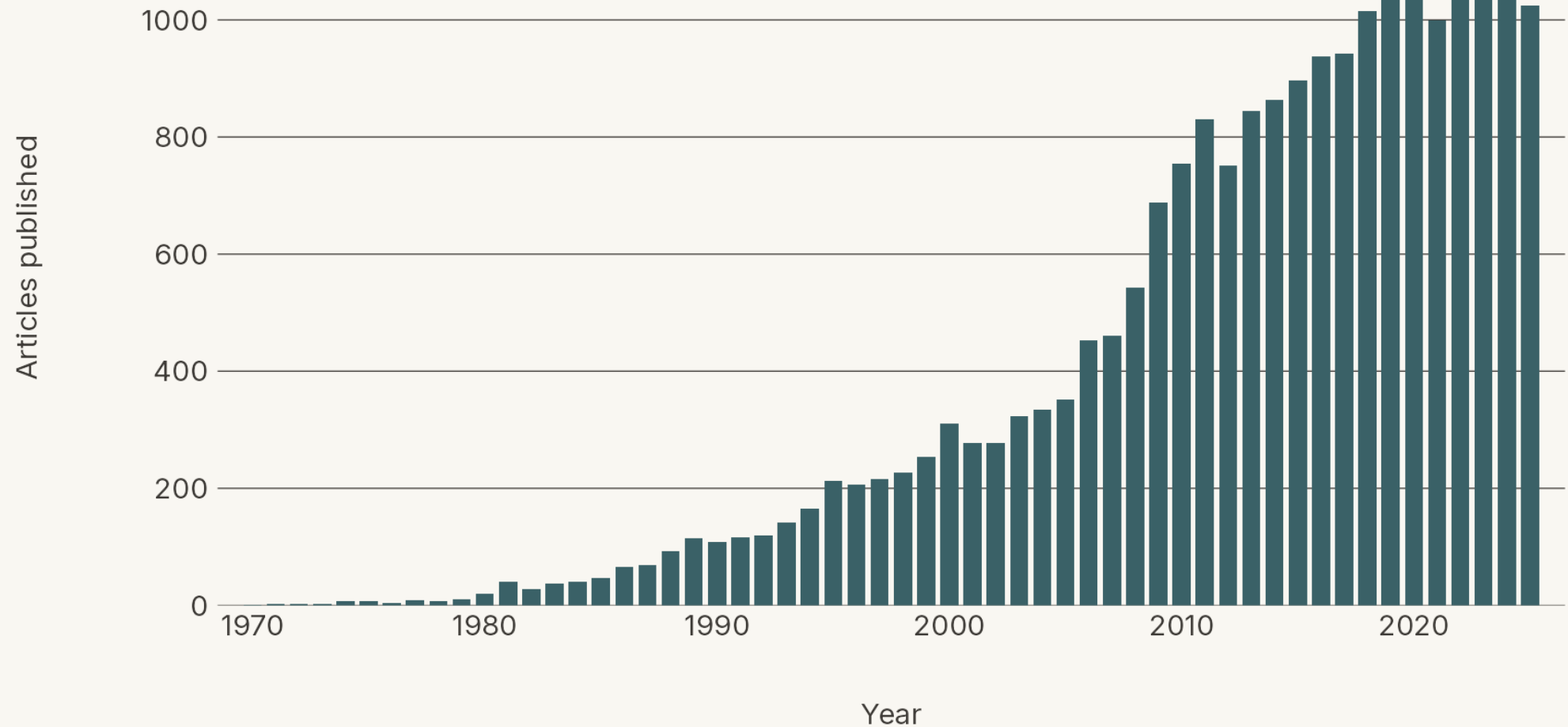


How quickly has the literature on organizational commitment grown?

Here's the count of articles published **per year**.

# Articles Published Per Year

Annual count of unique abstracts, 1970–2025 (n = 21,753 of 22,484)

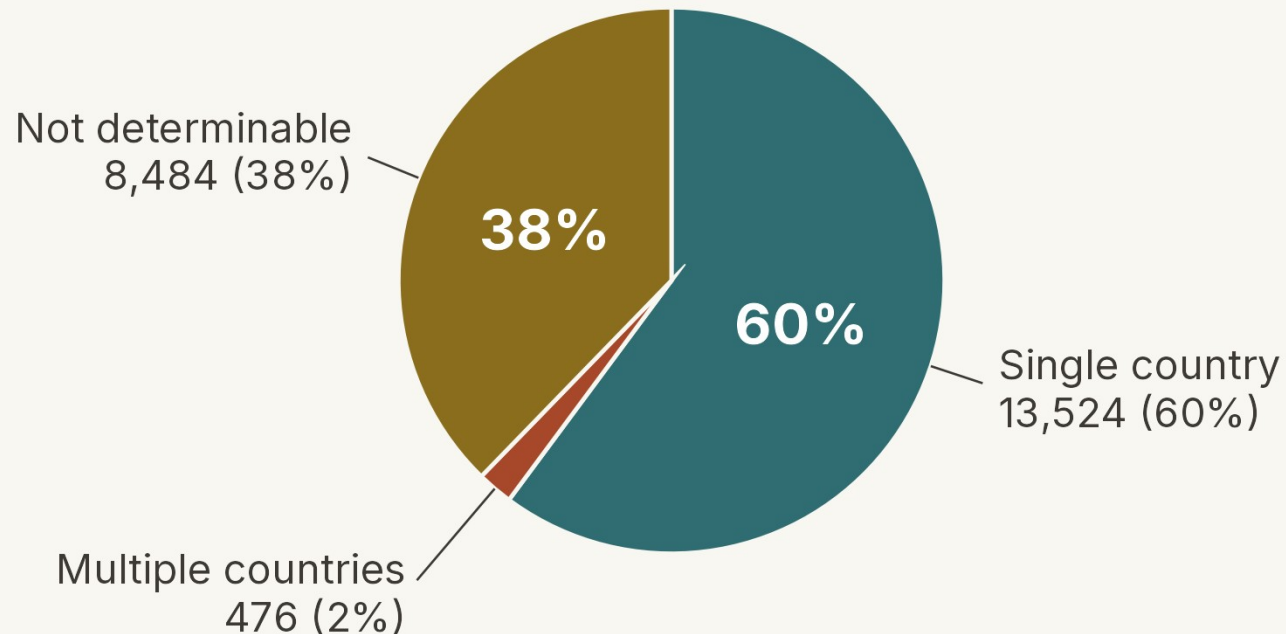




How has research on organizational commitment spread around the world over five decades?

# In what countries were the data obtained?

Country-coding status across all 22,484 unique abstracts



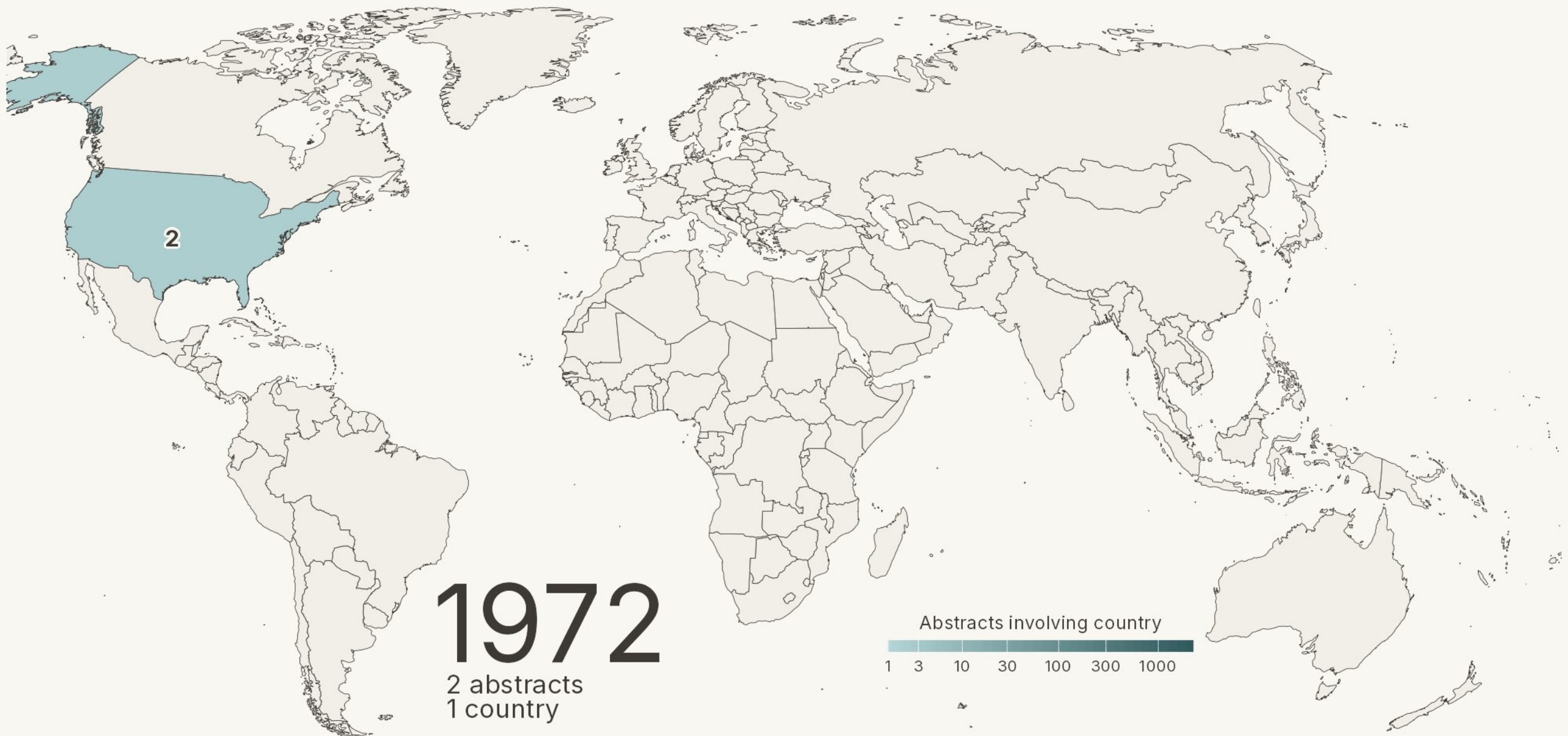
## How country of origin was determined

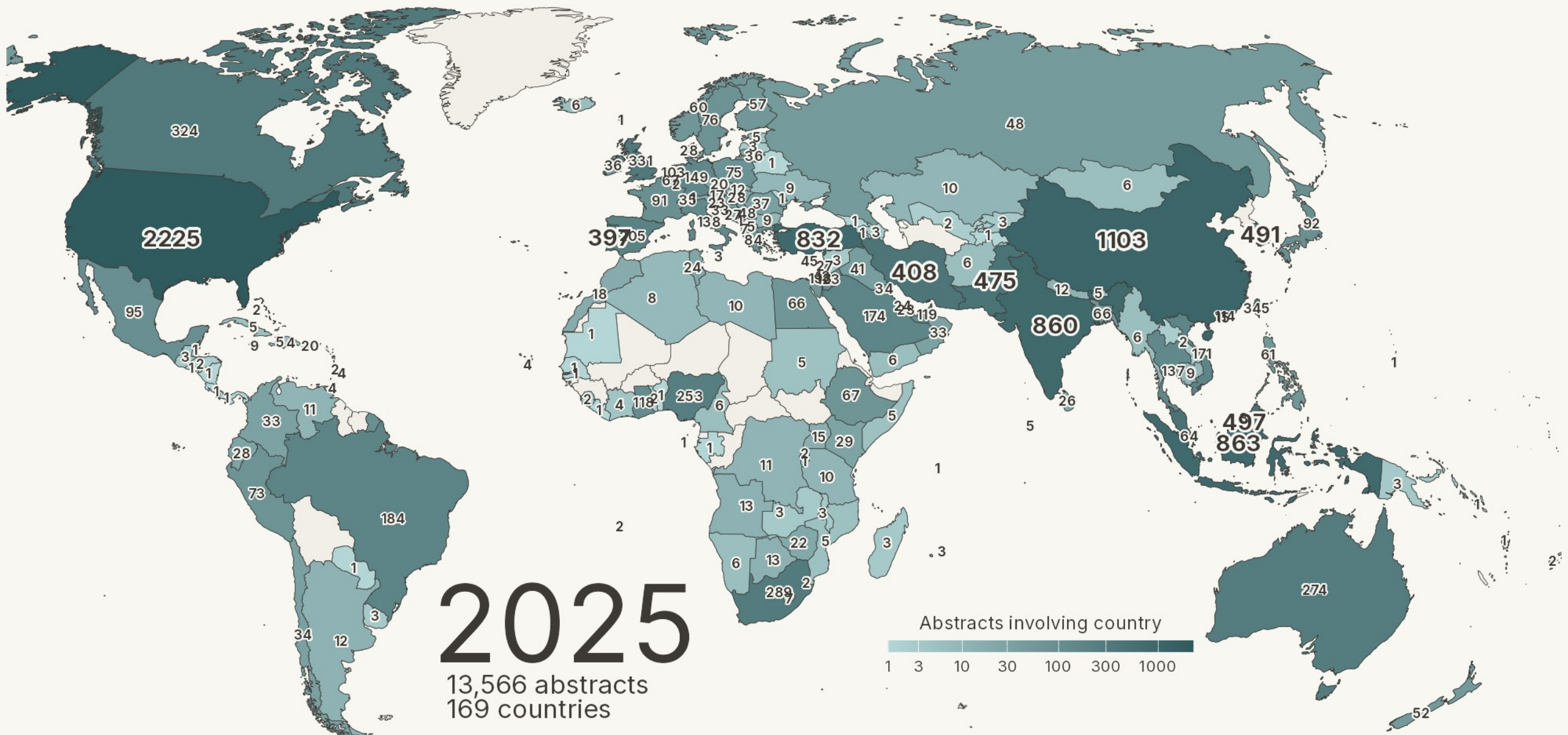
### Country coding criteria (each YES / NO):

- Is a country named directly in the abstract?
- Does a stated city or region map to a country?
- Does a named company or organization map to a country?
- Do all authors' affiliations point to the same country?

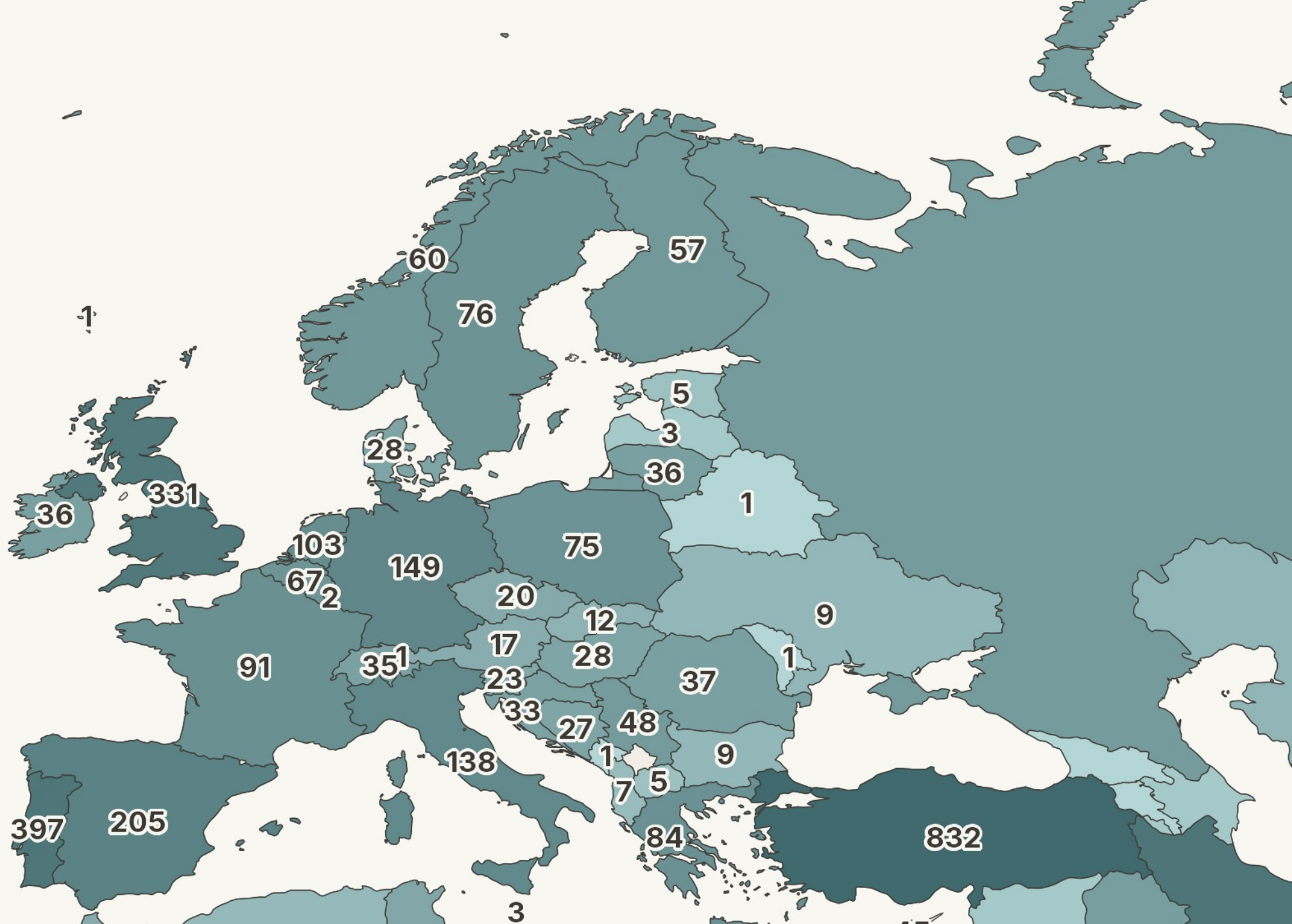
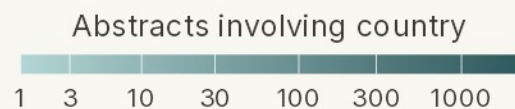
### Recovering undetermined cases:

- For undetermined cases, author affiliations were retrieved from OpenAlex via DOI lookup and passed to the model, resolving 14 previously undetermined items.



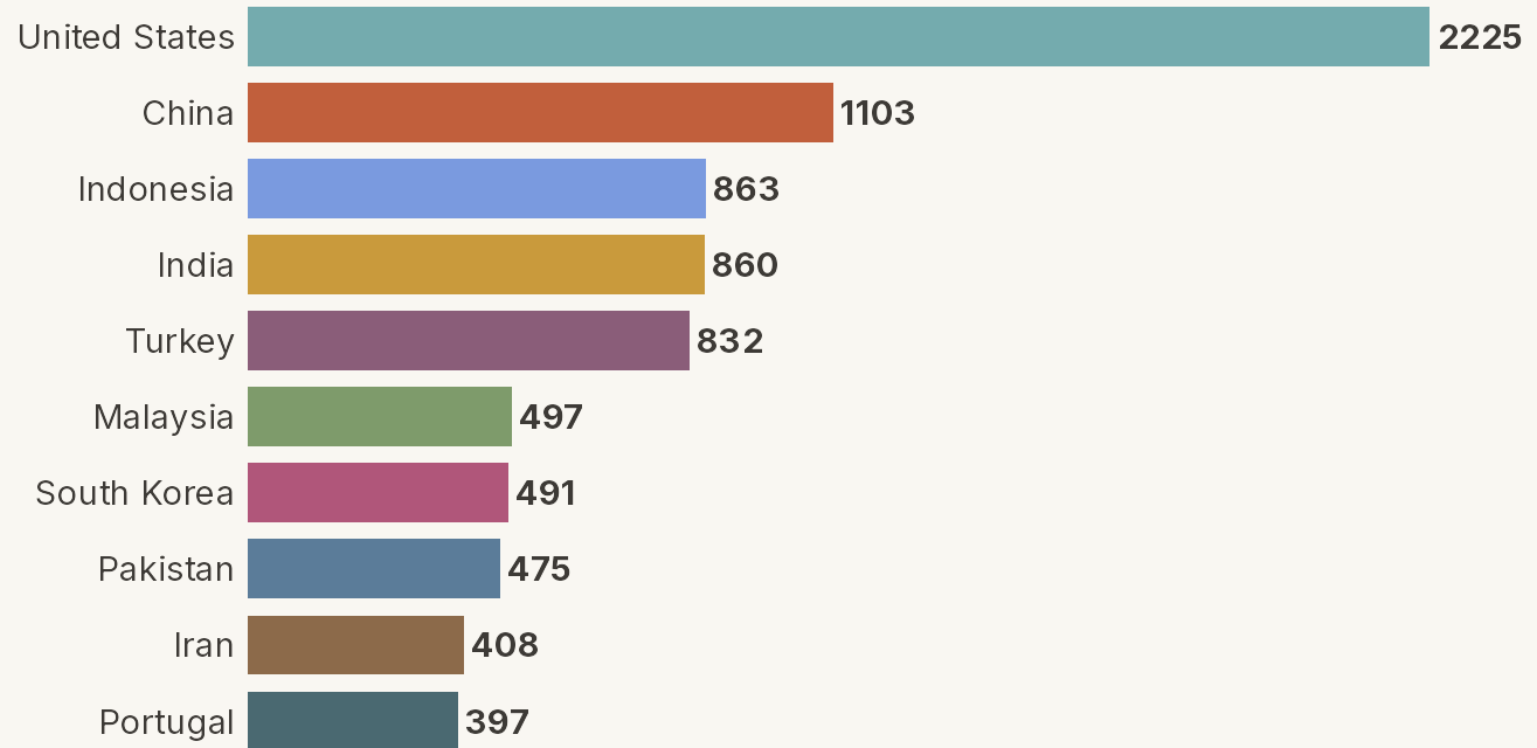


# 2025



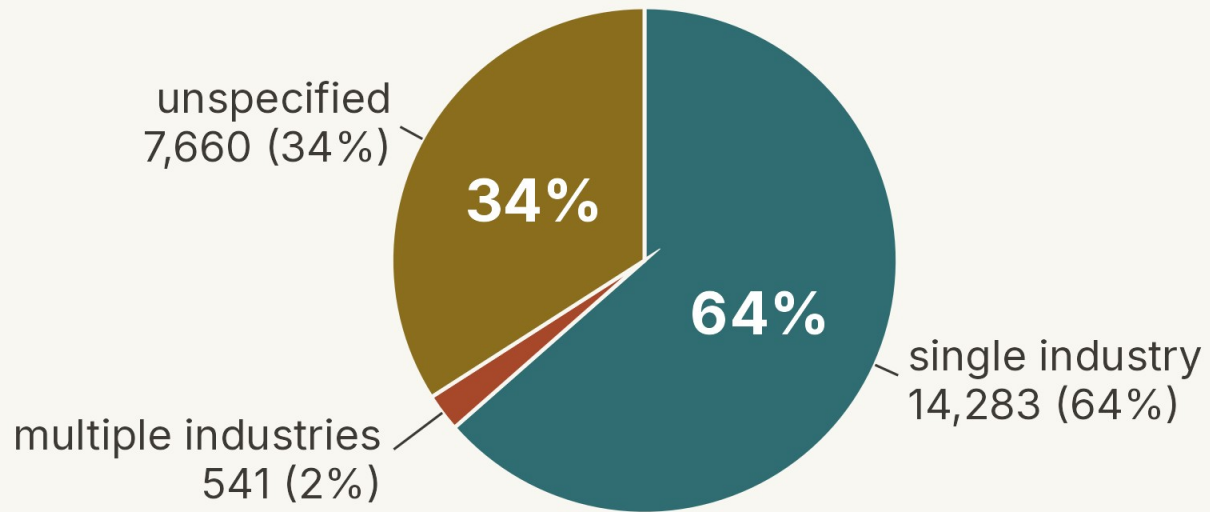
# Top 10 Countries by Cumulative Samples

# 2025



# What Industries have been Studied

Full coded sample, N = 22,484

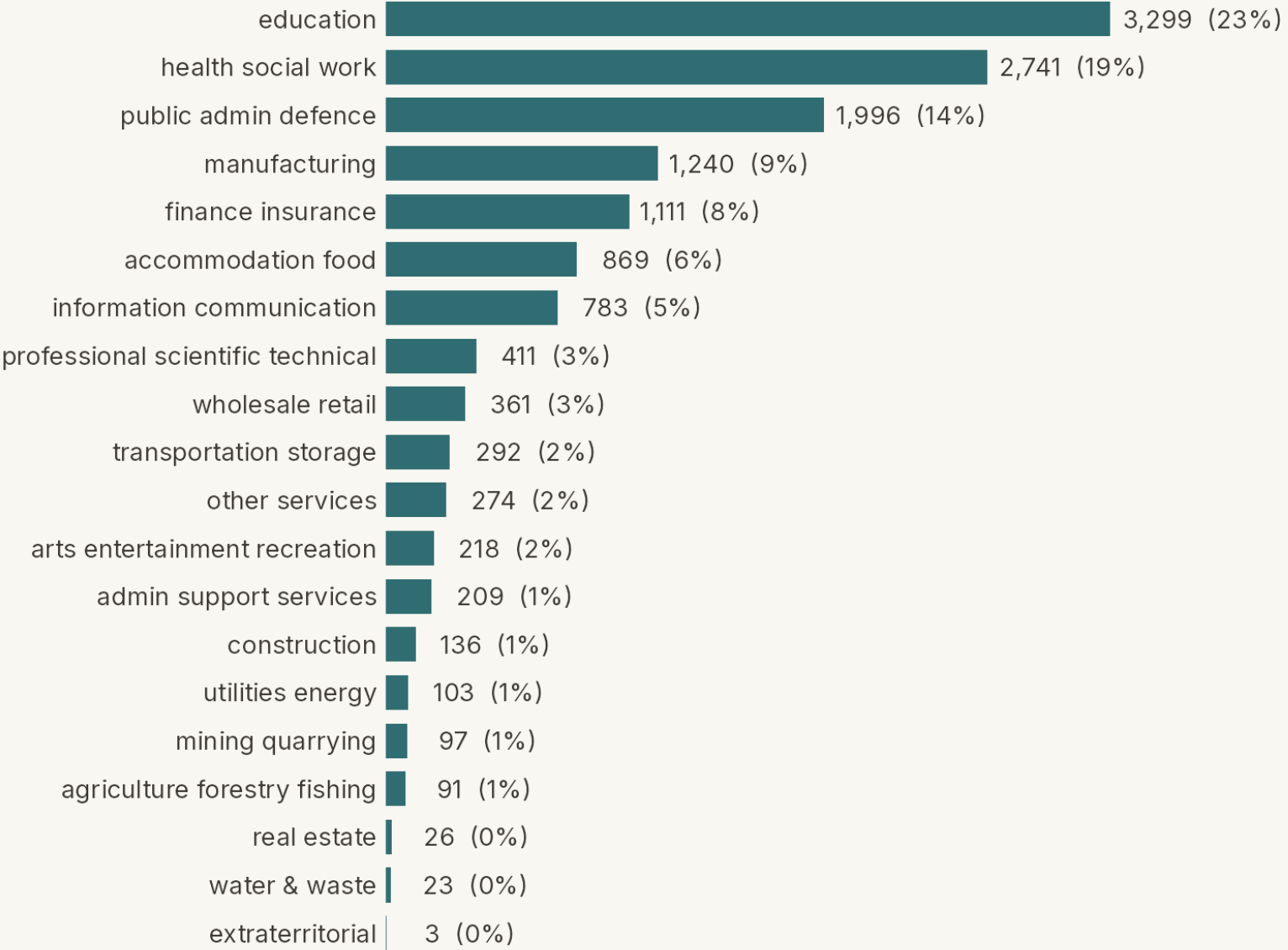


## Industry taxonomy — ISIC Rev. 4

The UN's standard industry taxonomy — 21 letter-coded sections, A (agriculture, forestry & fishing) through U (extraterritorial organizations).

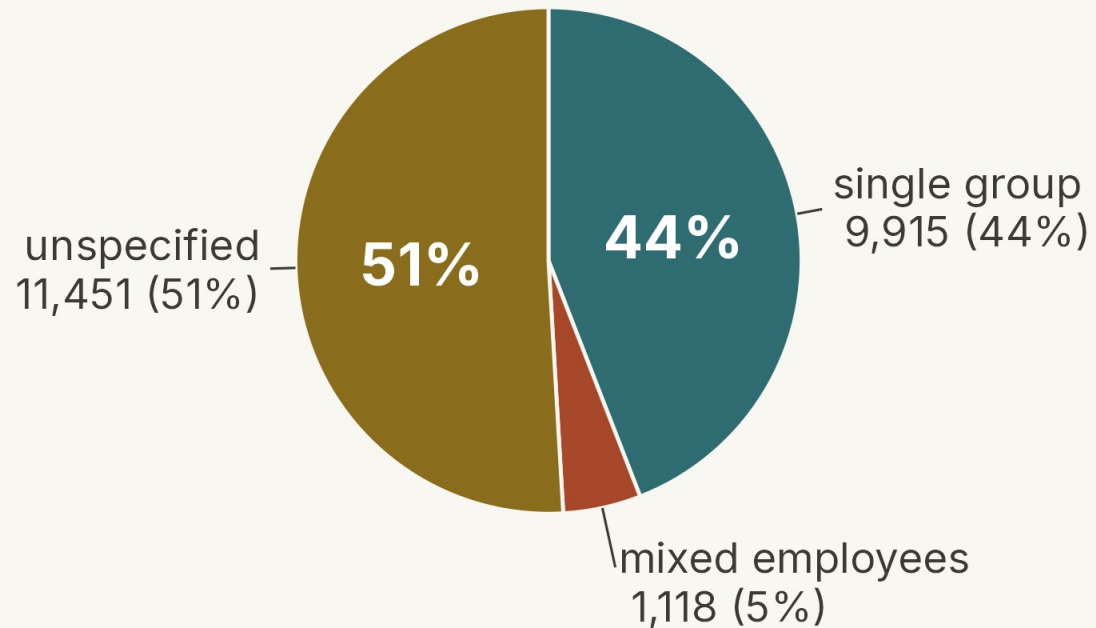
[Taxonomy Details Website](#)

Industry — distribution among single-industry studies



# What Occupations have been Studied

Full coded sample, N = 22,484



## Occupation taxonomy — ISCO-08

The ILO's standard occupation taxonomy (the counterpart to ISIC) — 10 major groups, 0 (armed forces) through 9 (elementary occupations).

[Taxonomy Details Website](#)

## Occupation — distribution among single-occupation studies



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How is organizational commitment positioned in the literature?

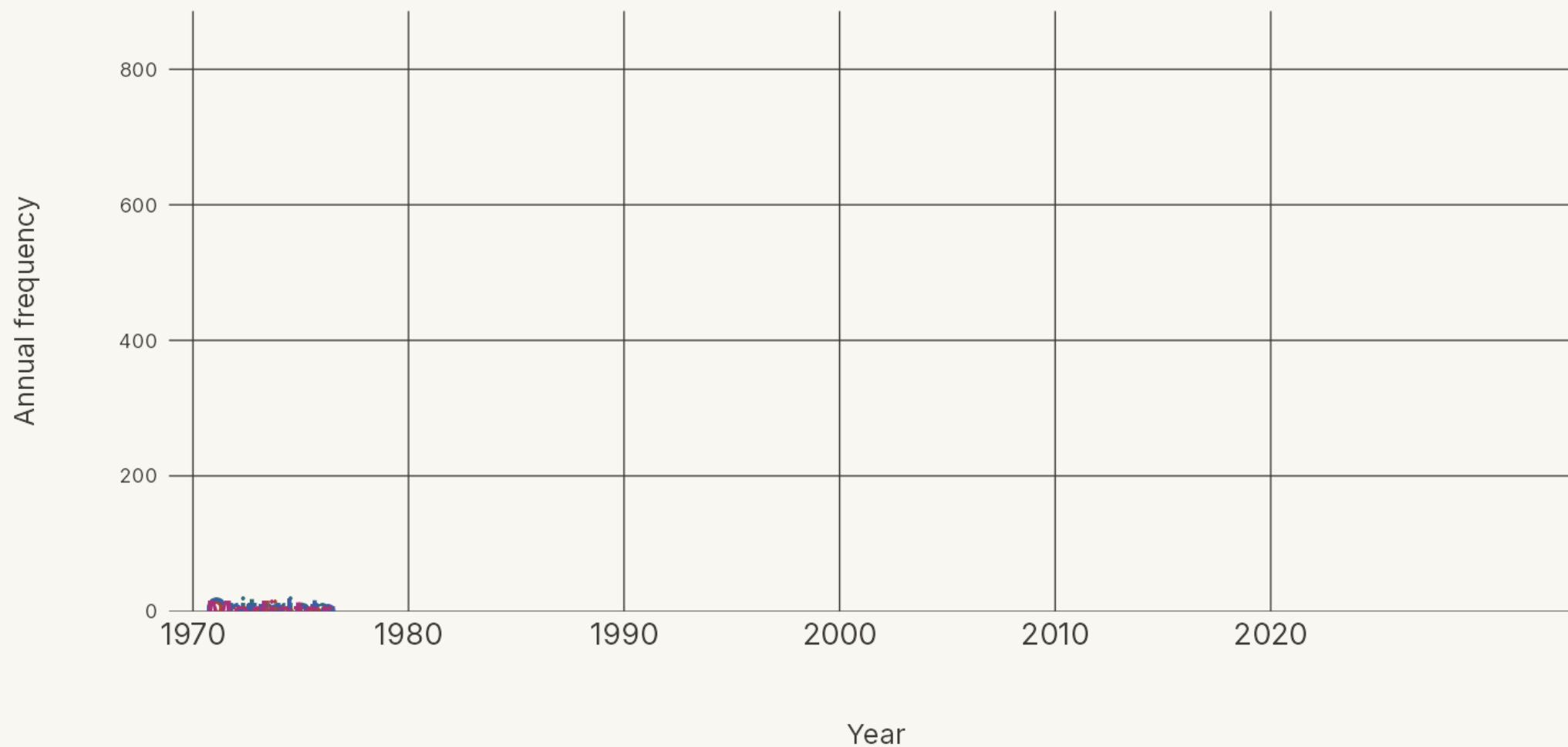
It plays four primary roles — as a predictor, a criterion, a correlate, and a mediator — and the balance has shifted over time.

**How the four roles were assigned**

We trained the AI to build a path model from each abstract's text. A variable's position in that path model determined which of the four roles it plays.

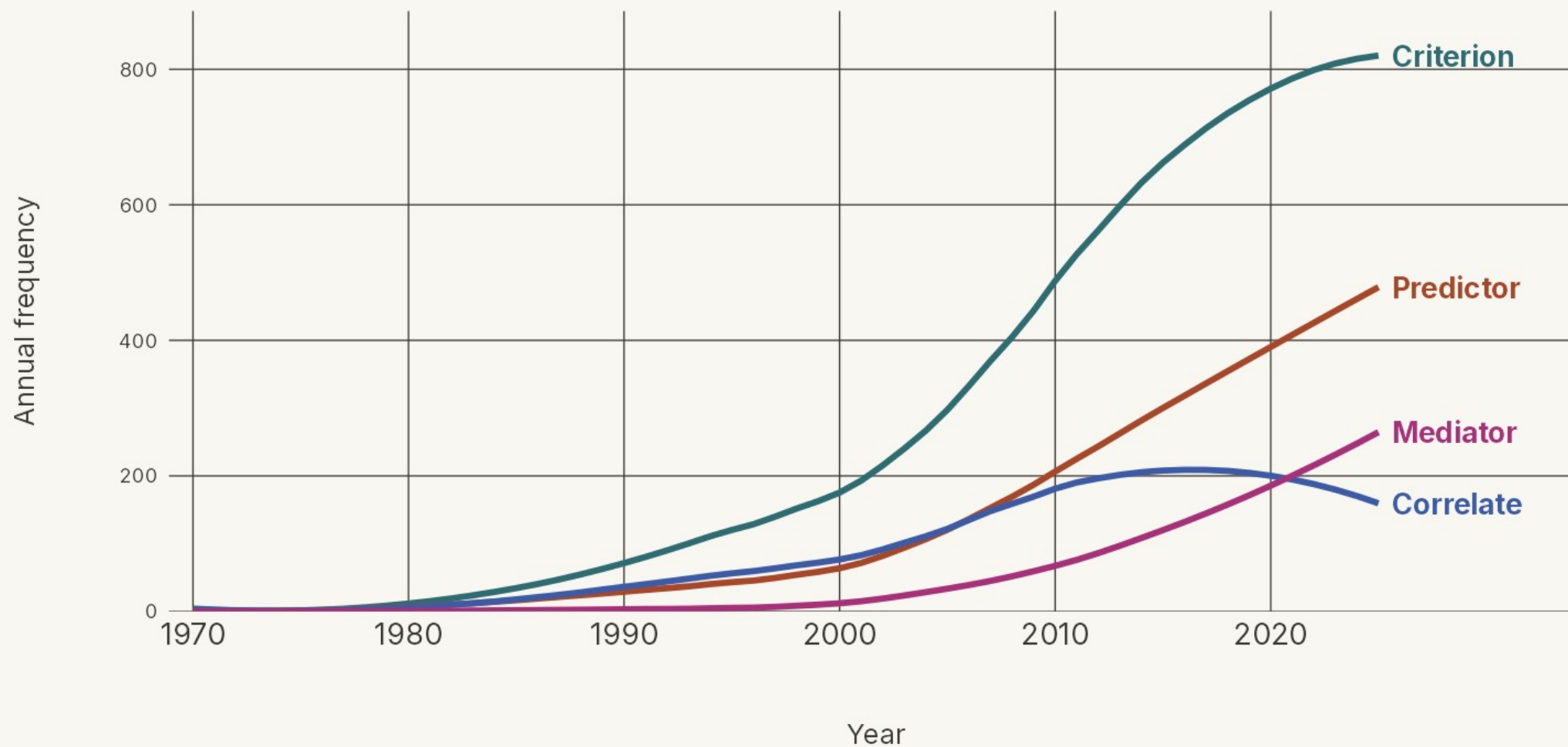
## How 'organizational commitment' is positioned

Annual count of abstracts using commitment in each role



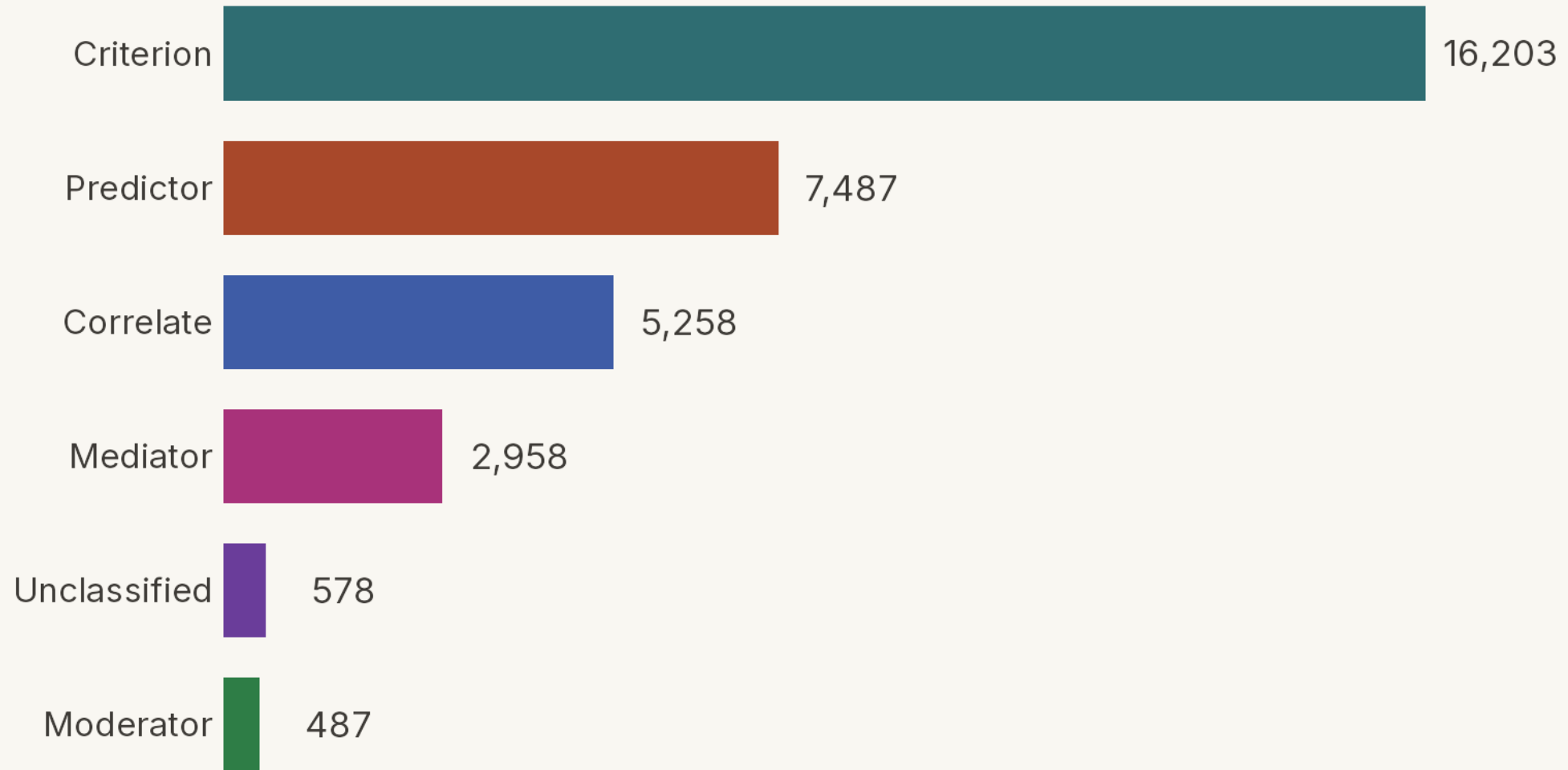
## How 'organizational commitment' is positioned

Annual count of abstracts using commitment in each role



## How 'organizational commitment' is positioned — all six roles

Total studies coded in each role — a study can hold several, so bars exceed the 22,484 coded



—

What variables served as predictors and criteria for commitment?

# How did we identify and combine variable labels

STEP

1

## Extract the raw labels

---

Direct extraction of the coded text,  
“as is.”

STEP

2

## Merge plural / spelling variants

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Combine simple plural and spelling variants (e.g., “policies” → “policy”).

STEP

3

## Group by meaning

---

**Predictors:** the AI proposed the families.

**Outcomes:** we supplied them; the AI sorted labels in.

Steps 2 and 3 by hand would be formidable tasks — AI makes them manageable.

# How many distinct variables did we extract?

After merging spelling and plural variants — before grouping by meaning

## Step 2

Merge plural / spelling variants

Predictors of commitment

**14,390**

distinct variables

Outcomes of commitment

**3,320**

distinct variables

# One construct, every language

For every study, the AI extracts the variables it examines — reading the abstract in its own language — and records each under one English label, so studies in any language pool into the same construct.

## ■ As written across languages

job satisfaction

工作满意度

satisfação no trabalho

iş tatmini

satisfacción laboral

satisfaction au travail

직무 만족

الرضا الوظيفي

English

Chinese

Portuguese

Turkish

Spanish

French

Korean

Arabic

## ■ The unified label

# job\_satisfaction

one English label · **2,085** studies pooled

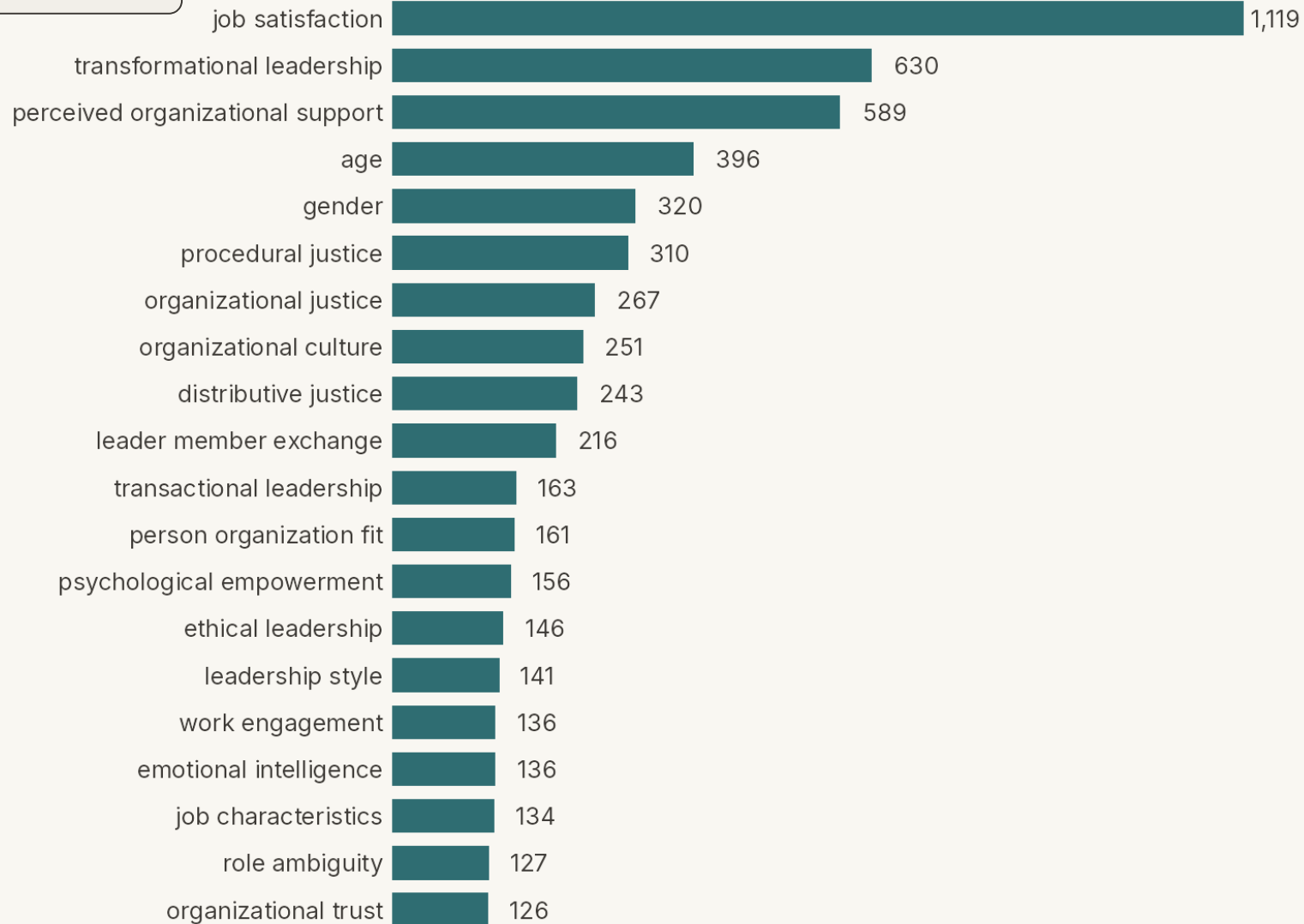
The alternative — human coding — would either **miss the non-English articles** or require **hiring multiple language-proficient coders**. The AI needs neither.

## Step 2

Merge plural / spelling variants

## Top 20 predictors of organizational commitment

Of 14,390 distinct predictor variables (light combining) — bar length = distinct studies naming each



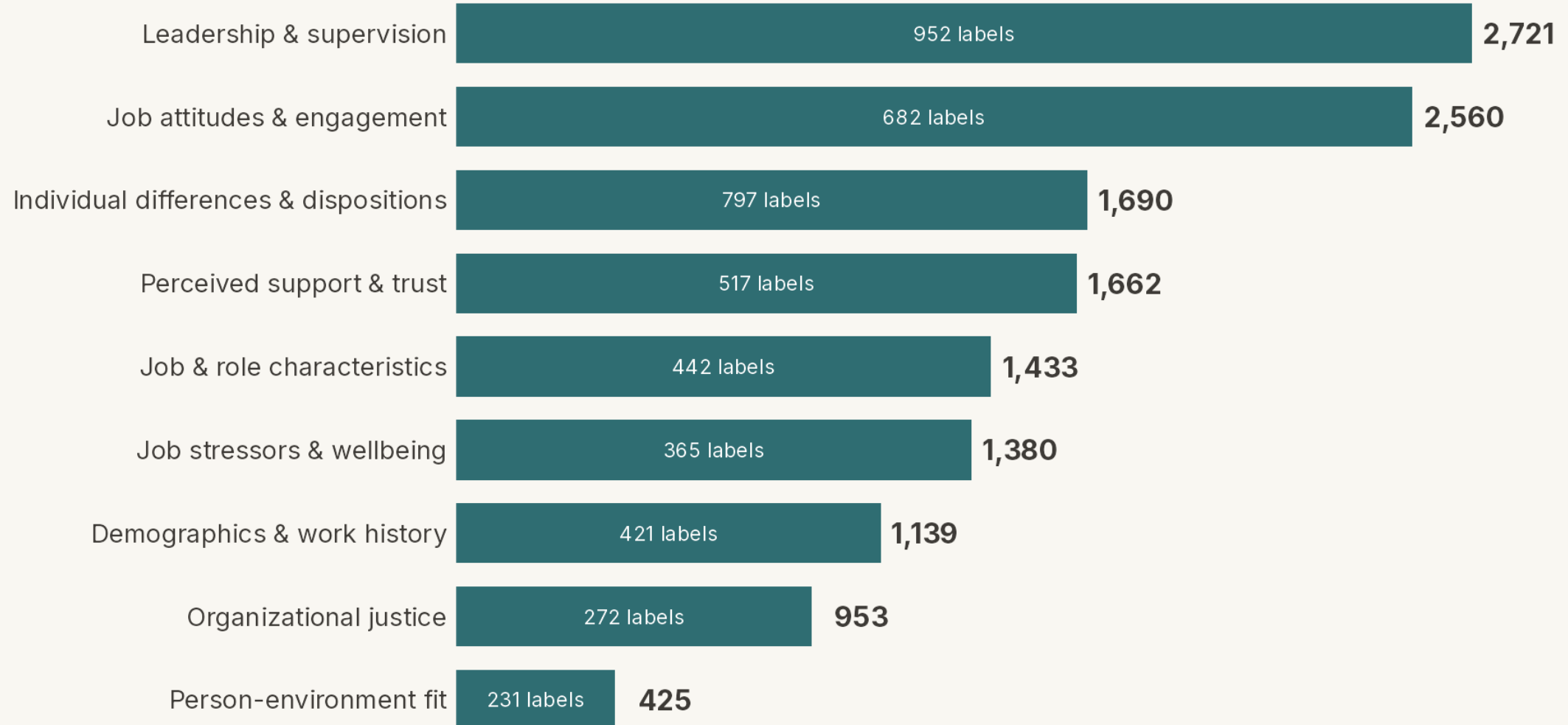
### Step 3

Group by meaning

## Predictors of organizational commitment, in nine categories

Bar & end label = distinct studies naming  $\geq 1$  predictor in the category.

Inside each bar = number of variable labels collapsed into it. ("Other" not shown.)

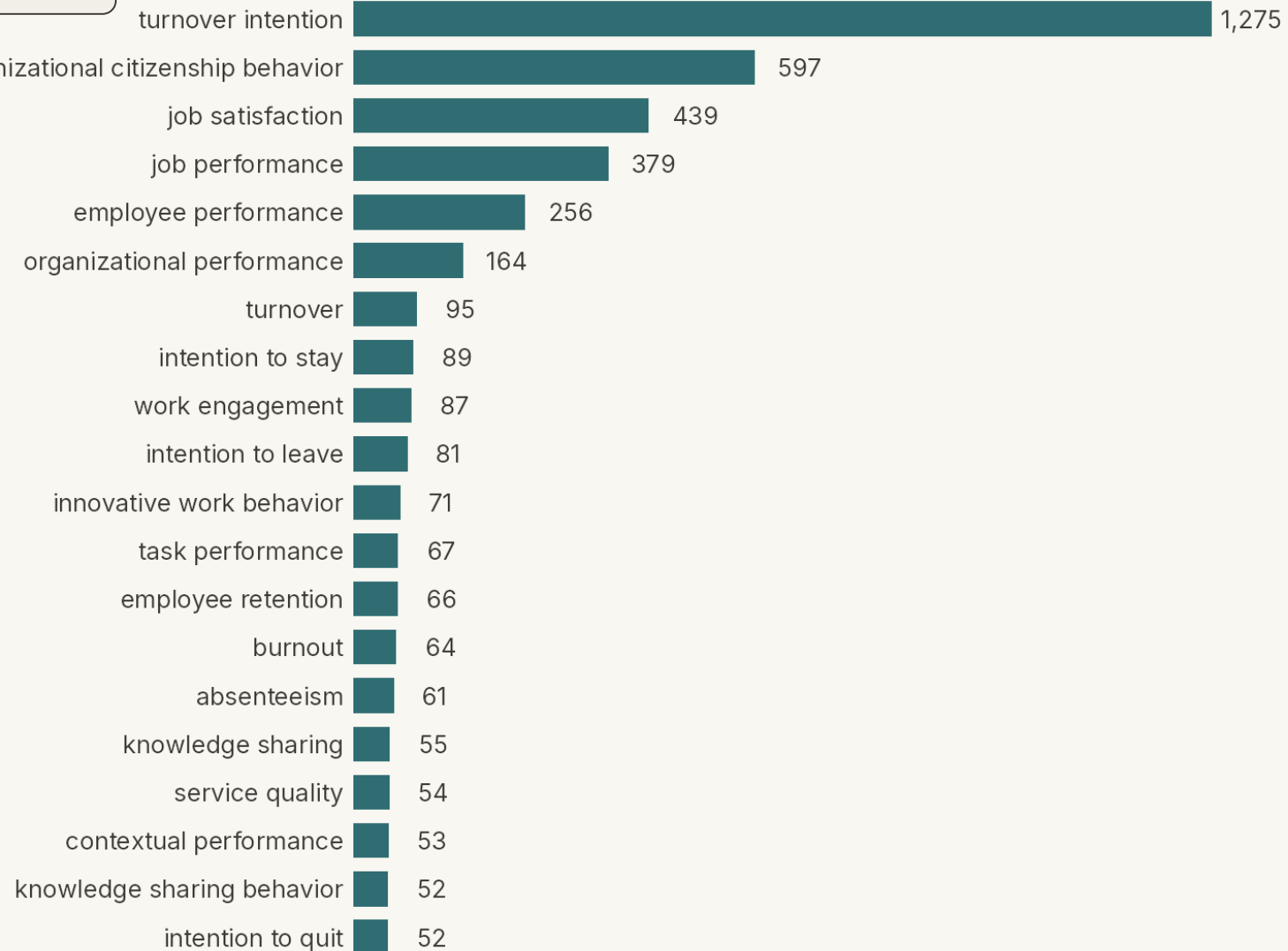


## Step 2

Merge plural / spelling variants

## Top 20 outcomes of organizational commitment

Of 3,320 distinct outcome variables (light combining) — bar length = distinct studies naming each



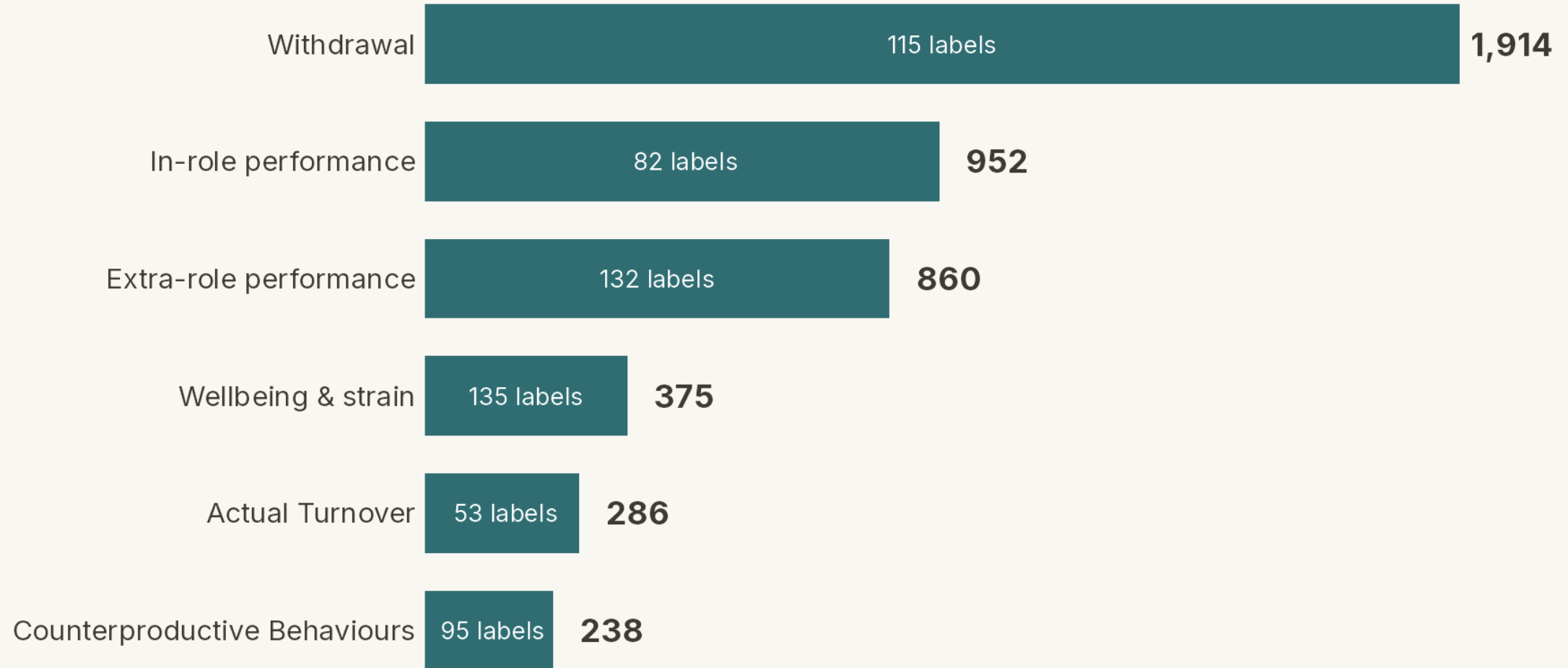
### Step 3

Group by meaning

## Outcomes of organizational commitment, in six categories

Bar & end label = distinct studies naming  $\geq 1$  outcome in the category.

Inside each bar = number of variable labels collapsed into it.



## Top 20 correlates of organizational commitment

Of 4,030 distinct correlate variables (light combining) — bar length = distinct studies naming each



FROM ABSTRACTS TO ANALYSIS

# *Next Steps*

---

The work ahead, in four steps.

# How accurate is AI coding? — from prior research

## ■ Relative Coding Accuracy

**97.0%** **92.1%**  
AI expert humans

of extracted values confirmed correct  
4 meta-analyses · 632 studies · 20,476 data points

## ■ Time saving with AI

**≈ 32×**

faster than coding by hand

130 hours of hand coding, done by the AI in about 4 hours.  
One meta-analysis: 12,025 numbers from 390 studies · one coder.

## ■ How the accuracy was measured

The AI and the human team each extracted every value independently. Matches were counted correct; disagreements were re-checked against the source article by an expert to decide which side was right. Each method's accuracy is the percentage of values confirmed correct, averaged across the four datasets.

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[Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. \(2026\). \*Improving Evidence Synthesis with Artificial Intelligence\* \[Preprint\].](#)

# Next steps

From the coded abstracts to a completed meta-analysis — four steps from here.

1

## Download the articles

Retrieve the full texts behind the coded abstracts.

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2

## Code the articles

Extract effect sizes and study details from each.

---

3

## Formalize the variable taxonomies

Finalize the construct groupings from these results.

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4

## Begin analyses

Run the meta-analytic models.



SCAN TO VIEW & SHARE THESE SLIDES

[tinyurl.com/icap2026-commitment](https://tinyurl.com/icap2026-commitment)

The download includes a Methodological Deep Dive we didn't have time to present.

# *Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature*

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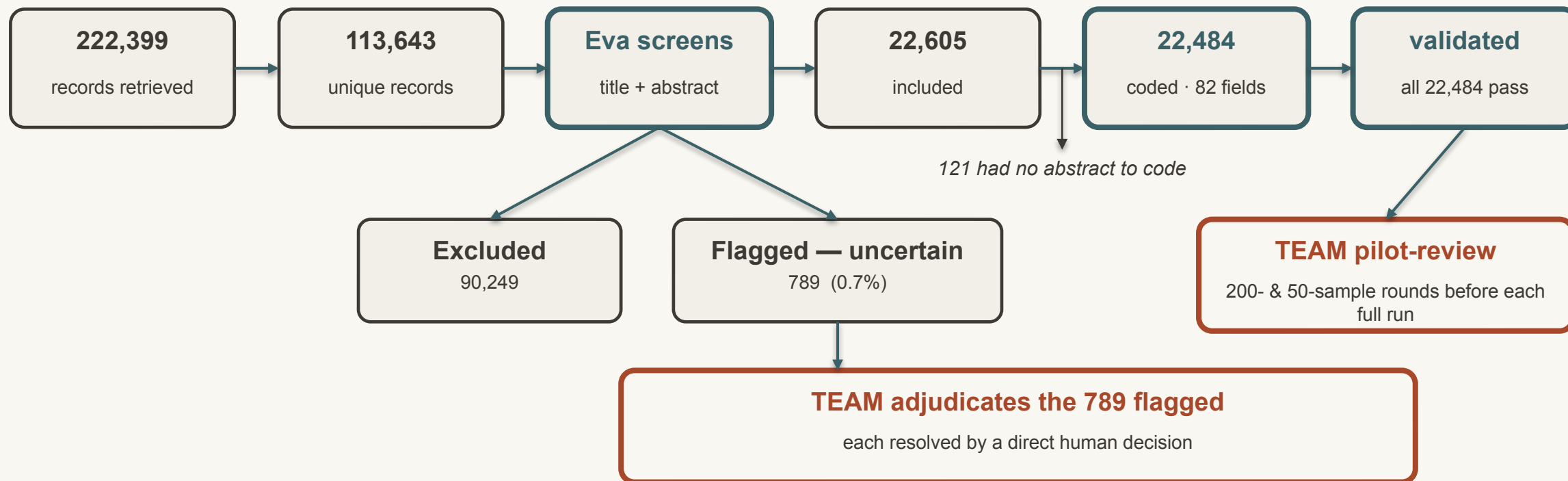
# *Methodological Deep Dive*

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The pipeline, calibration, coding scheme, and full counts — detail we skip in the talk.

# The workflow, end to end

Every record's path — the fan-out at screening, and the human decisions, made explicit



 automated (Eva)

 human (team)

 record counts

# The pipeline in full

Every record's path from retrieval to coded abstract — human steps marked in brown

|   |              |                                                                                                                                                                                                                          |
|---|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Retrieve     | 222,399 records — 7 databases, keyword + cited-reference searches                                                                                                                                                        |
| 2 | De-duplicate | 113,643 unique records after removing overlapping hits                                                                                                                                                                   |
| 3 | Screen       | Eva reads each record's title + abstract and applies 4 inclusion criteria — primary empirical, quantitative, organizational context, commitment measured: 22,605 include · 90,249 exclude · 789 flagged uncertain (0.7%) |
| 4 | Adjudicate   | Team resolves the 789 cases the AI flagged as uncertain — each gets a direct human decision (the AI never guesses on them)                                                                                               |
| 5 | Code         | 22,484 abstracts (121 of the 22,605 relevant records had no abstract to code) — 82 structured fields each, with a verbatim quote per field                                                                               |
| 6 | Validate     | All 22,484 pass automated consistency checks; team pilot-reviews before each full run                                                                                                                                    |

# Calibration & human oversight

Humans ran a blind spot-check, then concentrated effort on the cases the AI itself flagged as uncertain

| STAGE                      | WHEN         | WHAT HAPPENED                                                                                                                                                                                                                                                                     |
|----------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial alignment          | Sep 2025     | 40 abstracts blind-reviewed — 39/40 agreement (the one mismatch was a case Eva had flagged uncertain). Adopted the rule: when in doubt, err toward inclusion.                                                                                                                     |
| Uncertain-case calibration | Jan–Feb 2026 | 195 records from ~5,400 uncertain cases, blind-reviewed by three team members. Disagreements traced to undefined rules (missing / non-English abstracts) — the reviewers disagreed too. Rules made explicit; the whole set re-screened → uncertain cases fell from ~5,400 to 789. |
| Uncertainty adjudication   | Feb–Mar 2026 | Team hand-reviewed the 789 cases the AI flagged as uncertain — each resolved by a direct human decision rather than an AI guess.                                                                                                                                                  |
| Coding pilot               | May–Jun 2026 | Codebook built with the team; a 50-sample review + async re-test surfaced 5 fixes (e.g., a mandatory path-diagram step, a level-of-analysis field). Only then the full 22,484 run.                                                                                                |

Net: roughly 1% of the 113,643 records ever needed manual screening — concentrated exactly where the abstracts were genuinely ambiguous.

# What was coded

82 structured fields per abstract — each carrying a verbatim quote as its evidence

## PATH-MODEL FIRST

Eva drew each study's path model, located commitment within it, and only then assigned commitment's role — predictor, outcome, mediator, or moderator.

## The 82 fields capture

- Commitment role (predictor / outcome / mediator / moderator)
- Commitment type (affective / continuance / normative)
- Level of analysis (individual / unit / organization)
- Country of data collection (4-level evidence hierarchy)
- Industry — ISIC Rev. 4
- Occupation — ISCO-08
- Every predictor, outcome, mediator & moderator variable
- Variable names normalized to standard English (any source language)

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**Evidence, not assertion** — every field carries a verbatim quote from the abstract to support it.

# The numbers — and where they come from

Counts are verified from this project's database; accuracy figures belong to the validation study

## THIS PROJECT — DATABASE-VERIFIED

**222,399** records retrieved

**113,643** unique after de-duplication

**22,605** included · 90,249 excluded

**789** flagged uncertain — all human-adjudicated

**~1%** of records needed any manual screening

**22,484** abstracts coded · 82 fields each

**39/40** project blind-check agreement

## FROM THE VALIDATION STUDY — MEHR ET AL.

### Screening

97.2% / 91.6% vs expert human screeners

### Coding

97.0% / 92.1% vs expert human coders

### Tested on

5 meta-analyses · 30,150 screening documents

Sensitivity and specificity are properties of the validation study, not of this meta-analysis. This project reports only its own database-verified counts and the 39/40 blind-check.

SEED STUDIES FOR THE CITED-REFERENCE SEARCH

# *Foundational Works*

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What we mean by “foundational” — the studies whose citations we followed in the cited-reference search.

# Foundational works

1 OF 2

The cited-reference search retrieved every record citing any of these 14 works

**Allen, N. J., & Meyer, J. P. (1990).** The measurement and antecedents of affective, continuance and normative commitment to the organization. *Journal of Occupational Psychology*, 63(1), 1–18.

**Allen, N. J., & Meyer, J. P. (1996).** Affective, continuance, and normative commitment to the organization: An examination of construct validity. *Journal of Vocational Behavior*, 49(3), 252–276.

**Cook, J., & Wall, T. (1980).** New work attitude measures of trust, organizational commitment and personal need non-fulfilment. *Journal of Occupational Psychology*, 53(1), 39–52.

**Klein, H. J., Cooper, J. T., Molloy, J. C., & Swanson, J. A. (2014).** The assessment of commitment: Advantages of a unidimensional, target-free approach. *Journal of Applied Psychology*, 99(2), 222–238.

**Meyer, J. P., & Allen, N. J. (1984).** Testing the “side-bet theory” of organizational commitment: Some methodological considerations. *Journal of Applied Psychology*, 69(3), 372–378.

**Meyer, J. P., & Allen, N. J. (1991).** A three-component conceptualization of organizational commitment. *Human Resource Management Review*, 1(1), 61–89.

**Meyer, J. P., & Allen, N. J. (1997).** *Commitment in the workplace: Theory, research, and application*. Sage Publications.

# Foundational works

2 OF 2

The cited-reference search retrieved every record citing any of these 14 works

**Meyer, J. P., Allen, N. J., & Smith, C. A. (1993).** Commitment to organizations and occupations: Extension and test of a three-component conceptualization. *Journal of Applied Psychology*, 78(4), 538–551.

**Meyer, J. P., Stanley, D. J., Herscovitch, L., & Topolnytsky, L. (2002).** Affective, continuance, and normative commitment to the organization: A meta-analysis of antecedents, correlates, and consequences. *Journal of Vocational Behavior*, 61(1), 20–52.

**Mowday, R. T., Porter, L. W., & Steers, R. M. (1982).** *Employee–organization linkages: The psychology of commitment, absenteeism, and turnover*. Academic Press.

**Mowday, R. T., Steers, R. M., & Porter, L. W. (1979).** The measurement of organizational commitment. *Journal of Vocational Behavior*, 14(2), 224–247.

**O'Reilly, C. A., III, & Chatman, J. (1986).** Organizational commitment and psychological attachment: The effects of compliance, identification, and internalization on prosocial behavior. *Journal of Applied Psychology*, 71(3), 492–499.

**Porter, L. W., Steers, R. M., Mowday, R. T., & Boulian, P. V. (1974).** Organizational commitment, job satisfaction, and turnover among psychiatric technicians. *Journal of Applied Psychology*, 59(5), 603–609.

**Powell, D. M., & Meyer, J. P. (2004).** Side-bet theory and the three-component model of organizational commitment. *Journal of Vocational Behavior*, 65(1), 157–177.